

Christos Kalyvas-Kasopatidis, PhD

AI Security Architect · Vulnerability Intelligence

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PROFILE

I work where AI systems meet adversaries: securing the models, agents and pipelines organisations now depend on, and using AI to make cyber defence faster and more anticipatory. My background spans enterprise cyber defence, large-scale data engineering and a decade of academic research in machine learning and multi-objective optimisation.

In industry I have shaped AI security planning and controls, built AI asset visibility and shadow-AI detection, red-teamed LLM guardrails before adoption, and worked on the reliability and security of vulnerability intelligence platforms. I hold a PhD in Intelligent Big Data Management, an MSc in Information and Communication Systems Security and a BSc in Mathematics from the University of the Aegean, with 7 publications and about 150 citations.

INTERESTS

- Security of AI & LLM systems (prompt injection, guardrails, agentic AI)
- Vulnerability & threat intelligence
- AI for cyber defence
- Machine learning & multi-objective optimisation
- AI ethics & philosophy of mind
- Cognitive psychology of human–machine decision-making

EDUCATION

PhD in Intelligent Big Data Management 2013–2020
University of the Aegean
Dept. of Information and Communication Systems Engineering. Skyline queries and multi-objective (Pareto) optimisation combined with supervised machine learning over big data.

MSc in Information and Communication Systems Security 2010–2012
University of the Aegean
Thesis: formal modelling of cryptographic modes of operation with Alloy.

BSc in Mathematics 2005–2010
University of the Aegean
Pure mathematics, with coursework in cryptography, network security and secure system design.

FOCUS AREAS

Area	What I work on
Security of AI	AI security strategy and control design · prompt-injection and guardrail evaluation · agentic AI threat modelling · AI asset visibility and unsanctioned (shadow) AI detection
AI for security	Agentic workflows for security operations · ML detection engineering · AI-assisted code review and penetration-test reporting
Vulnerability & threat intelligence	Exploitation forecasting (KEV / EPSS) · advisory standards (CSAF) · CTI fusion with geopolitical and narrative signals
Decision intelligence	Multi-objective (Pareto / skyline) methods for ranking risk, choosing controls and briefing leadership under uncertainty

EXPERIENCE

AI Security Architect Sep 2021 – Present

Industry

- Shaped AI security planning: inventory → shadow-AI definition → visibility → maturity and risk roadmap
- Built a reproducible prompt-injection corpus mapped to OWASP LLM01 and MITRE ATLAS to evaluate commercial guardrails before adoption
- Built an agentic AI prototype for security operations and designed agent-to-SIEM patterns for automated alert triage and enrichment
- Designed a user–asset–data-classification model joining network, endpoint and cloud telemetry into governance and risk metrics
- ML detection engineering: time series, positive-unlabelled learning and pattern mining for C2 beaconing, DGA domains and compromised hosts
- Production incident response and root-cause analysis, monitoring design, penetration-test cycles and Kubernetes platform decisions
- Trained defender teams to design their own prompts and agents for daily operations

Teacher Jan 2021 – Jul 2021

Directorate of Secondary Education, Kavala

Taught first-grade secondary school classes.

Software Engineer (research) Apr 2019 – Apr 2020

University of Piraeus

- Built a prototype index and query engine for spatio-textual big data
- Prototyped dimensionality reduction combining clustering with hierarchical graph partitioning over spatial data and co-occurrence graphs

Platform Engineer Oct 2019 – Jan 2020

University of the Aegean

- Designed and deployed a data-analytics infrastructure and data fabric for research teams, and presented it to project partners

Computer Programmer & Analyst / System Administrator (military service) Apr 2018 – Dec 2018

Hellenic Army

- Deployed a medium-scale network with Active Directory Domain Services in under a day; set up backup, disaster recovery and user support

Solution Architect & System Administrator Jan 2017 – Jan 2020

University of the Aegean

- Deployed a scalable compute cluster at zero additional cost from open-source tools and idle hardware
- Enforced access control, encryption and auditing; ran data migrations with minimal downtime

Lab Technical Lead Oct 2013 – Jan 2020

University of the Aegean

- Led labs in object-oriented programming, databases, data structures and discrete mathematics; mentored students

Software Engineer & System Administrator Oct 2013 – Oct 2015

University of the Aegean

- Built a marine GIS routing system optimising vessel routes for fuel, environmental impact and risk
- Designed the database architecture, optimised retrieval queries and automated testing with Selenium

RESEARCH PROJECTS

Publicly funded R&D projects (Greece and EU) I contributed to as a researcher and engineer:

- **ENIRISST**: Intelligent Research Infrastructure for Shipping, Supply Chain, Transport and Logistics. Data-analytics infrastructure and data fabric for the University of the Aegean research teams.
- **SPADES**: SPAtio-textual Data Exploration at Scale. Indexing and query processing for spatio-textual big data (University of Piraeus).

- **AMINESS**: Analysis of Marine Information for Environmentally Safe Shipping. Marine GIS and route optimisation for fuel use, environmental impact and voyage risk (University of the Aegean).

TALKS & RECOGNITION

- **Endorser, Generative Application Firewall (GAF)** (2026). Acknowledged as an endorser of the GAF research on a unified security layer for LLM applications and AI agents, alongside researchers from Cambridge, MIT CSAIL and the OWASP GenAI Security Project. Paper (arXiv:2601.15824) · Announcement
- **Guest, Career Con: "Career prospects in cybersecurity"** (2025, in Greek). Career guidance talk on cybersecurity, AI security and data science. Watch on YouTube

SKILLS

AI Security

AI security governance
AI security plans and controls · OWASP LLM Top 10 · MITRE ATLAS · NIST AI RMF · EU AI Act

LLM red-teaming
Prompt-injection testing · guardrail evaluation · agentic AI threat modelling

AI for security
Agentic SOC use cases · LLM-assisted code scanning · LangChain / LangGraph · Azure AI Foundry

Security & Intelligence

Vulnerability intelligence
CVE · KEV · EPSS · CSAF · coordinated vulnerability disclosure

Threat intelligence & detection
MITRE ATT&CK · STIX/TAXII · OpenCTI · MISP · Splunk · Chronicle · Sentinel

Governance & regulation
NIS2 · CRA · GDPR · ISO 27001 · NIST CSF · risk assessment

Data & Engineering

Machine learning
Time series · PU learning · pattern mining · TensorFlow · PyTorch · SparkML

Data & retrieval
Python · SQL · Neo4j · OpenSearch · Qdrant · PostgreSQL · Hadoop / Spark

SRE & cloud
Kubernetes · Docker · Linux · WAF · monitoring · incident response · GitHub Actions

FRAMEWORKS & COMMUNITY

Frameworks: OWASP LLM Top 10 · MITRE ATLAS · MITRE ATT&CK · NIST AI RMF · EU AI Act · NIS2 · Cyber Resilience Act · ISO/IEC 27001

Community: OWASP AI Exchange · FIRST (AI SIG) · Hellenic Artificial Intelligence Society (EETN)

Expert registries: European Commission expert database, EU Funding & Tenders Portal (registered expert since 2025) · National Registry of Certified Evaluators, General Secretariat for Research and Innovation, Greece (Law 4310/2014)

CERTIFICATIONS

- Agentic Artificial Intelligence for Leaders · Maven · 2025
- AI Techniques in Cybersecurity · National & Kapodistrian University of Athens · 2024
- Chronicle SIEM Fundamentals · Google · 2023
- Splunk Core Certified Power User · Splunk · 2021
- Introduction to Quantum Computing · Qubit by Qubit / IBM Quantum · 2021

PUBLICATIONS

Journal articles

1. Kalyvas, C., Manolis Maragoudakis (2020). **A skyline-based decision boundary estimation method for binominal classification in big data**. *Computation*, 8(3), 80. doi:10.3390/computation8030080
2. Kalyvas, C., Manolis Maragoudakis (2019). **Skyline and reverse skyline query processing in SpatialHadoop**. *Data & Knowledge Engineering*, 122, 55–80. doi:10.1016/j.datak.2019.04.004
3. Kalyvas, C., Athanasios Kokkos, Theodoros Tzouramanis (2017). **A survey of official online sources of high-quality free-of-charge geospatial data for maritime geographic information systems applications**. *Information Systems*, 65, 36–51. doi:10.1016/j.is.2016.11.002

Conference papers

1. Kalyvas, C., Manolis Maragoudakis (2020). **A skyline-based decision boundary estimation method for binominal classification in big data**. In *2020 5th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)*, pp. 1–8, IEEE. doi:10.1109/SEEDA-CECNSM49515.2020.9221822
2. Kalyvas, C., Theodoros Tzouramanis, Yannis Manolopoulos (2017). **Processing skyline queries in temporal databases**. In *Proceedings of the Symposium on Applied Computing (SAC 2017)*, pp. 893–899, ACM. doi:10.1145/3019612.3019677
3. Kalyvas, C., Elisavet Konstantinou, Georgios Kambourakis (2012). **Modeling multiple modes of operation with Alloy**. In *Computer Applications for Security, Control and System Engineering* (CCIS 339), pp. 78–85, Springer. doi:10.1007/978-3-642-35264-5_11

Preprints

1. Kalyvas, C., Theodoros Tzouramanis (2017). **A survey of skyline query processing**. *arXiv preprint* arXiv:1704.01788.